Table 1: PVC pipe – summary statistics of the raw measurements

Measurement item	Mean	Std. dev.	Type A unc.	Type B unc.	Combined unc.
Inner diameter (cm)	2.02300	0.00377	0.000843	0.00144	0.00167
Outer diameter (cm)	2.60825	0.00373	0.000833	0.00144	0.00167
Length (cm)	4.10225	0.00697	0.00156	0.00144	0.00212
Mass (g)	12.51000	0.00459	0.00103	0.00289	0.00306
Buoyancy reading (gw)	8.69600	0.0266	0.00596	0.00289	0.00662

Table 2: Cylinder – summary statistics of the raw measurements

Measurement item	Mean	Std. dev.	Type A unc.	Type B unc.	Combined unc.
Diameter (cm)	1.19500	0.00459	0.00103	0.00144	0.00177
Height (cm)	4.99375	0.00425	0.000951	0.00144	0.00173
Mass (g)	6.75600	0.00598	0.00134	0.00289	0.00318

Table 3: Steel ball – summary statistics of the raw measurements

Measurement item	Mean	Std. dev.	Type A unc.	Type B unc.	Combined unc.
Diameter (mm)	11.88395	0.00507	0.00113	0.00289	0.0031
Mass (g)	6.87450	0.00605	0.00135	0.00289	0.00319

Table 4: PVC pipe – density calculation

Quantity	Value
Inner radius (cm)	1.01150 ± 0.00084
Outer radius (cm)	1.30412 ± 0.00084
Inner circle area (cm^2)	3.2143 ± 0.0054
Outer circle area (cm^2)	5.3430 ± 0.0069
Base area (cm^2)	2.1288 ± 0.0087
Volume (cm^3)	8.733 ± 0.036
Density (g/cm^3)	1.4325 ± 0.0059
Density via Archimedes' principle (g/cm^3)	1.4386 ± 0.0012

Table 5: Cylinder – density calculation

Quantity	Value
Radius (cm)	0.59750 ± 0.00089
Base area (cm^2)	1.1216 ± 0.0034
Volume (cm^3)	5.601 ± 0.017
Density (g/cm^3)	1.2062 ± 0.0037

Table 6: Steel ball – density calculation

Quantity	Value
Radius (mm)	5.9420 ± 0.0016
Volume (cm^3)	0.87878 ± 0.00069
Density (g/cm^3)	7.8228 ± 0.0072

Table 7: Results summary – PVC pipe

Quantity	Value
Inner diameter (cm)	2.0230 ± 0.0017
Outer diameter (cm)	2.6082 ± 0.0017
Length (cm)	4.1022 ± 0.0022
Mass (g)	12.5100 ± 0.0031
Density (g/cm^3)	1.4325 ± 0.0059
Buoyancy reading (gw)	8.6960 ± 0.0067
Density via Archimedes' principle (g/cm^3)	1.4386 ± 0.0012

Table 8: Results summary – Cylinder

Quantity	Value
Diameter (cm)	1.1950 ± 0.0018
Height (cm)	4.9938 ± 0.0018
Mass (g)	6.7560 ± 0.0032
Density (g/cm^3)	1.2062 ± 0.0037

Table 9: Results summary – Steel ball

Quantity	Value
Diameter (mm)	11.8839 ± 0.0031
Mass (g)	6.8745 ± 0.0032
Density (g/cm^3)	7.8228 ± 0.0072

Table 10: Raw measurements – PVC pipe

Inner diameter (cm)	Outer diameter (cm)	Length (cm)	Mass (g)	Buoyancy reading (gw)
2.025	2.600	4.100	12.51	8.70
2.025	2.615	4.100	12.51	8.68
2.020	2.610	4.100	12.50	8.67
2.030	2.605	4.100	12.51	8.66
2.020	2.610	4.110	12.50	8.71
2.020	2.610	4.100	12.51	8.64
2.020	2.605	4.095	12.51	8.66
2.020	2.610	4.095	12.51	8.69
2.025	2.615	4.125	12.51	8.69
2.020	2.605	4.100	12.51	8.68
2.015	2.610	4.100	12.52	8.69
2.025	2.605	4.100	12.51	8.69
2.020	2.610	4.105	12.51	8.69
2.020	2.605	4.100	12.51	8.74
2.025	2.610	4.115	12.51	8.71
2.030	2.610	4.100	12.51	8.72
2.025	2.605	4.100	12.52	8.72
2.025	2.610	4.100	12.51	8.72
2.025	2.610	4.100	12.51	8.73
2.025	2.605	4.100	12.51	8.73

Table 11: Raw measurements – Cylinder

Diameter (cm)	Height (cm)	Mass (g)
1.195	4.995	6.76
1.185	4.990	6.75
1.195	4.990	6.76
1.195	4.995	6.75
1.195	5.005	6.76
1.205	4.995	6.76
1.200	4.990	6.75
1.195	4.990	6.76
1.195	4.990	6.75
1.195	4.995	6.75
1.195	5.000	6.75
1.195	4.995	6.77
1.200	4.990	6.76
1.190	4.990	6.76
1.190	4.990	6.75
1.195	4.995	6.76
1.200	4.995	6.76
1.190	4.990	6.76
1.200	5.000	6.75
1.190	4.995	6.75

Table 12: Raw measurements – Steel ball

Zero reading (mm)	Diameter reading (mm)	Corrected diameter (mm)	Mass (g)
0.000	11.872	11.872	6.87
0.000	11.881	11.881	6.88
0.000	11.888	11.888	6.88
0.000	11.887	11.887	6.88
-0.002	11.876	11.878	6.88
-0.002	11.879	11.881	6.88
-0.002	11.883	11.885	6.87
-0.002	11.891	11.893	6.88
0.000	11.892	11.892	6.87
0.000	11.881	11.881	6.88
-0.002	11.876	11.878	6.88
-0.002	11.882	11.884	6.86
-0.002	11.880	11.882	6.87
-0.002	11.878	11.880	6.87
-0.002	11.882	11.884	6.87
-0.002	11.883	11.885	6.88
-0.002	11.883	11.885	6.87
-0.002	11.886	11.888	6.88
-0.002	11.883	11.885	6.87
-0.002	11.888	11.890	6.87